

Algebra: The Distance Formula

Practice Worksheet • numberbender.com



Name: _____

Date: _____

Score: ____ / 10

DIRECTIONS

Use the Distance Formula $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$ to find the distance between each pair of points. Simplify all radicals.

1. Find d:

(0, 0) and (3, 4)

Answer: _____

2. Find d:

(1, 2) and (4, 6)

Answer: _____

3. Find d:

(-3, 0) and (0, 4)

Answer: _____

4. Find d:

(2, -1) and (2, 5)

Answer: _____

5. Find d:

(-1, -1) and (2, 3)

Answer: _____

6. Find d:

(0, -3) and (4, 0)

Answer: _____

7. Find d:

(-2, 3) and (4, -5)

Answer: _____

8. Find d:

(3, -2) and (-1, 1)

Answer: _____

9. Find d:

(-5, 2) and (3, -4)

Answer: _____

10. Which is farther? A: (0,0)→(6,8) or B: (-3,1)→(3,-7)

Show work for both.

Answer: _____

Based on the Numberbender lesson "ALGEBRA: Using the Distance Formula" • youtu.be/5JilGerPyW4



Answer Key & Solutions

Algebra: The Distance Formula • Numberbender

TEACHER NOTES

Items 1–6: results are perfect squares — no radical simplification needed. Items 7–8: results simplify cleanly. Items 9–10: compare or apply distances. Remind students to square both differences before adding under the radical.

1. Find d: (0, 0) and (3, 4)

Answer: 5

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

2. Find d: (1, 2) and (4, 6)

Answer: 5

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

3. Find d: (–3, 0) and (0, 4)

Answer: 5

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

4. Find d: (2, –1) and (2, 5)

Answer: 6

$$\sqrt{(0^2+6^2)} = \sqrt{36} = 6.$$

5. Find d: (–1, –1) and (2, 3)

Answer: 5

$$\sqrt{(3^2+4^2)} = \sqrt{(9+16)} = \sqrt{25} = 5.$$

6. Find d: (0, –3) and (4, 0)

Answer: 5

$$\sqrt{(4^2+3^2)} = \sqrt{(16+9)} = \sqrt{25} = 5.$$

7. Find d: (–2, 3) and (4, –5)

Answer: 10

$$\sqrt{(6^2+8^2)} = \sqrt{(36+64)} = \sqrt{100} = 10.$$

8. Find d: (3, –2) and (–1, 1)

Answer: 5

$$\sqrt{(4^2+3^2)} = \sqrt{(16+9)} = \sqrt{25} = 5.$$

9. Find d: (–5, 2) and (3, –4)

Answer: 10

$$\sqrt{(8^2+6^2)} = \sqrt{(64+36)} = \sqrt{100} = 10.$$

10. Which is farther? A: (0,0)→(6,8) or B: (–3,1)→(3,–7) Show work for both.

Answer: Equal — both d = 10

A: $\sqrt{(36+64)}=10$; B: $\sqrt{(36+64)}=10$. Same distance.

